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Phase-change materials for non-volatile photonic applications

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Phase-change materials (PCMs) provide a unique combination of properties. On transformation from the amorphous to crystalline state, their optical properties change drastically. Short optical or electrical pulses can be utilized to switch between these states, making PCMs attractive for photonic applications. We review recent developments in PCMs and evaluate the potential for all-photonic memories. Towards this goal, the progress and existing challenges to realize waveguides with stepwise adjustable transmission are presented. Colour-rendering and nanopixel displays form another interesting application. Finally, nanophotonic applications based on plasmonic nanostructures are introduced. They provide reconfigurable, non-volatile functionality enabling manipulation and control of light. Requirements and perspectives to successfully implement PCMs in emerging areas of photonics are discussed.

Optical devices to guide, manipulate and detect light are key to modern photonic systems. They provide information transport through waveguides and optical fibres, and enable displays, memories and integrated optical sensors. Interestingly, electronic systems also realize similar functionalities, where electrons rather than photons are utilized to carry or store information, to name just two functions. Recent decades have seen an exciting competition between optical and electronic solutions to realize these functionalities. In both electronics and photonics, remarkable progress and exponential growth has been achieved for many properties, such as storage density (Moore's law) for semiconductor devices[1] and the transmission rate of optical fibres (Keck's law)[2]. Today, photons are preferable if information has to be transported, due to the much higher bandwidth of optical systems compared with electronic systems. Yet, electronic systems have already reached a very high level of integration and miniaturization, giving them an edge when data has to be stored. They also have an efficient transistor-based logic. Hence, electronic systems are currently superior in data processing and storage, as they offer a 'programmable' flexibility in terms of electronic 're-wiring' for different purposes. Thus, for the next generation of competitive photonic systems, a higher level of integration and miniaturization, as well as adjustable, on-demand functionalities, is required to realize novel photonic devices. For all-optical switching or routers without moving mechanical parts, materials that provide suitable, adjustable optical properties are required.

In this Review, we illustrate the role of PCMs in emerging photonic applications, ranging from optoelectronic systems capable of multi-level storage and display towards their integration in nanophotonic systems and metamaterials, which will enable new functionalities on demand. Before doing so, it is necessary to provide a clear definition of the term 'phase-change material'. At present, this term is utilized in different communities with three different meanings. PCMs have been discussed since the 1980s for energy storage[3]. In this case, the ability of these materials for latent heat storage is employed. The history of PCMs for optical applications starts considerably earlier, as can be seen from ref. 4. This paper already mentions fast phase change (crystallization and amorphization) on laser irradiation, which is accompanied by a pronounced non-volatile change of optical properties.

In recent years, the term 'phase-change material' has unfortunately also been used in a broader sense, for example, in a review paper[5], to include oxides, such as vanadium dioxide, as well[6,7]. In this and other materials, a temperature-induced change of optical properties is observed. This phase transition is usually reversible: when the material is cooled down again below the transition temperature, without special sample treatment[8], it returns to the original state at this temperature. Hence, these 'phase-transition materials' cannot be directly utilized where reconfigurable, non-volatile applications are desired.

We also leave out interesting recent work where the optical properties only change on the (volatile) response of charge carriers, for example, via pulsed optical excitation[9] or electrical voltages[10]. We instead focus on the class of non-volatile PCMs and their application potential for non-volatile photonic applications, but will omit the word 'non-volatile' in most cases in the following.

Basic material properties

PCMs have a striking portfolio of properties (Box 1). They can be rapidly and reversibly switched between amorphous and crystalline states. Yet, the optical and electronic properties of the amorphous and crystalline states differ tremendously. The ability to switch rapidly between two states with different properties qualifies these materials for applications in data storage. Indeed, the first commercially successful application of PCMs was realized in rewritable optical data storage. In such storage systems, a short pulse of a focused laser beam with high intensity locally heats the PCM above its melting temperature. Rapidly quenching the phase-change alloy with cooling rates higher than 10^9 K s^{-1} freezes the liquid-like state into a disordered, amorphous phase. Such huge cooling rates can only be obtained if small amounts of material are heated by very short pulses above the melting temperature and subsequently the thermal excitation is turned off, so that the heated region of material is rapidly quenched. The resulting amorphous state has different optical properties than the surrounding crystalline state. Hence, the detection of regions with amorphous structure is straightforward using a laser beam of low intensity. To erase the stored information, a laser pulse with intermediate power is utilized.

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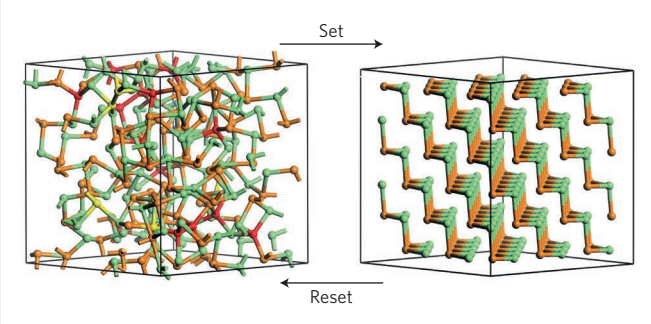
Box 1 | Resonant bonding.

PCMs, as employed in data storage applications and photonics, utilize a remarkable property: the pronounced change of the dielectric function $\varepsilon(\omega)$ on crystallization. In solids, usually the amorphous and crystalline states have very similar optical properties. This can be seen when comparing glass (amorphous SiO_2) with a quartz crystal (crystalline SiO_2). Both solids have a very similar dielectric function. Small differences in the dielectric function are mainly due to the lower density of amorphous SiO_2 and subtle differences in the density of electronic states near the energy gap of the amorphous state. W. H. Zachariasen had already explained this similarity in 1932, when he argued that oxide glasses have the same atomic arrangement as the corresponding crystals, since they are held together by the same forces[86]. In the language of modern chemistry, we might say that both states are characterized by the same bonding mechanism. In PCMs, on the contrary, the dielectric function of the amorphous and crystalline states often differs significantly. This drastic change is related to a pronounced change in chemical bonding on the transition from the amorphous to the crystalline state. Schematically, this is displayed in the figure, where the atomic arrangement in crystalline and amorphous GeTe, a typical PCM, is depicted (Ge atoms (3-fold) orange; Ge atoms (4-fold) red/yellow; Te atoms, green). Crystalline GeTe is characterized by a rhombohedral arrangement, where bond angles are close to 90°, leading to rather similar nearest-neighbour (r_{short}) and next-nearest-neighbour (r_{long}) spacings (r_{long} / r_{short} = 1.1), so that each atom has an average coordination number[87] of almost 6. In amorphous GeTe, the majority of the atoms are also in an octahedral-like atomic arrangement, but now the bond angles deviate more strongly from 90° and the difference between the first- and second-nearest neighbours has increased (r_{long} / r_{short} = 1.27)[88]. In the amorphous state of GeTe, ordinary covalent bonds are formed. Each atom has on average three nearest neighbours and ordinary covalent bonds provide the interaction between adjacent atoms. In crystalline PCMs, the number of valence electrons per atom (three) is insufficient to saturate all six bonds. Hence, the solid exhibits resonance bonding, where two adjacent atoms are held together by a single electron. This bonding mechanism leads to high electronic polarizabilities as evidenced by a large optical dielectric constant ε_∞ (refs 21,85,89). For the applications discussed here, a number of aspects are important. How pronounced is the contrast between the amorphous and the crystalline states? The answer is shown in Fig. 1, where the dielectric function of different PCMs is shown for the amorphous and crystalline states. The largest contrast, that is, difference in the dielectric function, is observed below 1 eV, that is, in the infrared. Here, the refractive index on crystallization more than doubles. At the same time, PCMs in this energy range also have a non-negligible absorption, a finding that has to be considered in the design of photonic devices.



In addition, the question arises how rapidly the material can switch back and forth between the two different states. Crystallization is slower than amorphization, hence crystallization is the time-limiting step. Nevertheless, the fastest crystallization processes observed in PCMs at elevated temperatures proceed within a few nanoseconds or even less. Apparently, the kinetics of the crystallization process are strongly influenced by the unconventional temperature dependence of the crystal growth velocity. At high temperatures, the activation energy for crystal growth is very low, while the activation barrier increases by up to an order of magnitude at lower temperatures[90,91]. This pronounced temperature dependence explains the excellent stability of amorphous marks against crystallization at room temperature and above, while the low activation barrier at around 800 K explains the ease of re-crystallization and hence the high switching speeds. At present, it is not yet fully clear how this temperature dependence relates to the unique bonding mechanism utilized in crystalline PCMs.

This pulse locally heats the phase-change film above the crystallization temperature. At elevated temperatures above the glass transition temperature, the atoms become increasingly mobile and can revert to the energetically favourable crystalline state, erasing the recorded information. So far, three different formats have been realized to store audio, video and high-definition video with rewriting capabilities (compact disk, digital versatile disk and Blu-ray disk). These different formats employ an increasingly shorter laser wavelength and have enabled realization of a minimum mark length of 149 nm (ref. 11). While optical data storage was a viable option in the late 1980s and 1990s, the rapid decrease of memory cell size in electronic memories has increasingly undermined the relevance and market share of optical storage solutions. This is mainly because it is rather demanding to realize optical storage concepts that break the diffraction limit[12].

Interestingly, in recent years, PCMs have also been investigated intensely for application in non-volatile electronic memories[13,14], where the metastability of the amorphous and crystalline states is used to store data. Here, a short voltage pulse is used to rapidly and reversibly switch the PCM in a confined cell from the amorphous to the crystalline state. Impressively, small cell sizes of 7.5 nm × 17.5 nm have already been realized and switching has even been demonstrated for clusters down to 2 nm in diameter[1,15,16]. These findings demonstrate the excellent scaling potential of PCMs in non-volatile memories. In addition, the switching process is very fast, with switching times in the nanosecond or even subnanosecond regime[2,17,18]. These attractive attributes even leave room to hope that PCMs might be suited to realize a universal memory, which combines the attributes of flash and dynamic random-access memories[11,19,20].

In recent years, the ability to switch rapidly and reversibly between two states with very different optical properties has found an increasingly broader variety of potential applications in photonics, including optoelectronic storage and display systems. For multi-level storage, the pronounced optical contrast between the two states is utilized in conjunction with the ability to control the state of transformation locally, as discussed in more detail later. To unravel the potential of these applications, we present typical properties of PCMs, which also elucidate in which energy (wavelength) range the optical contrast is particularly pronounced, where it comes from, which materials show it and how fast we can switch.

Figure 1 shows the refractive index n and the extinction coefficient k for an energy range up to 2.5 eV (wavelength of 0.5 μm). All PCMs displayed in Fig. 1 are characterized by a pronounced increase in the refractive index on crystallization. Typically, the refractive index below the bandgap, below which interband transitions cannot be

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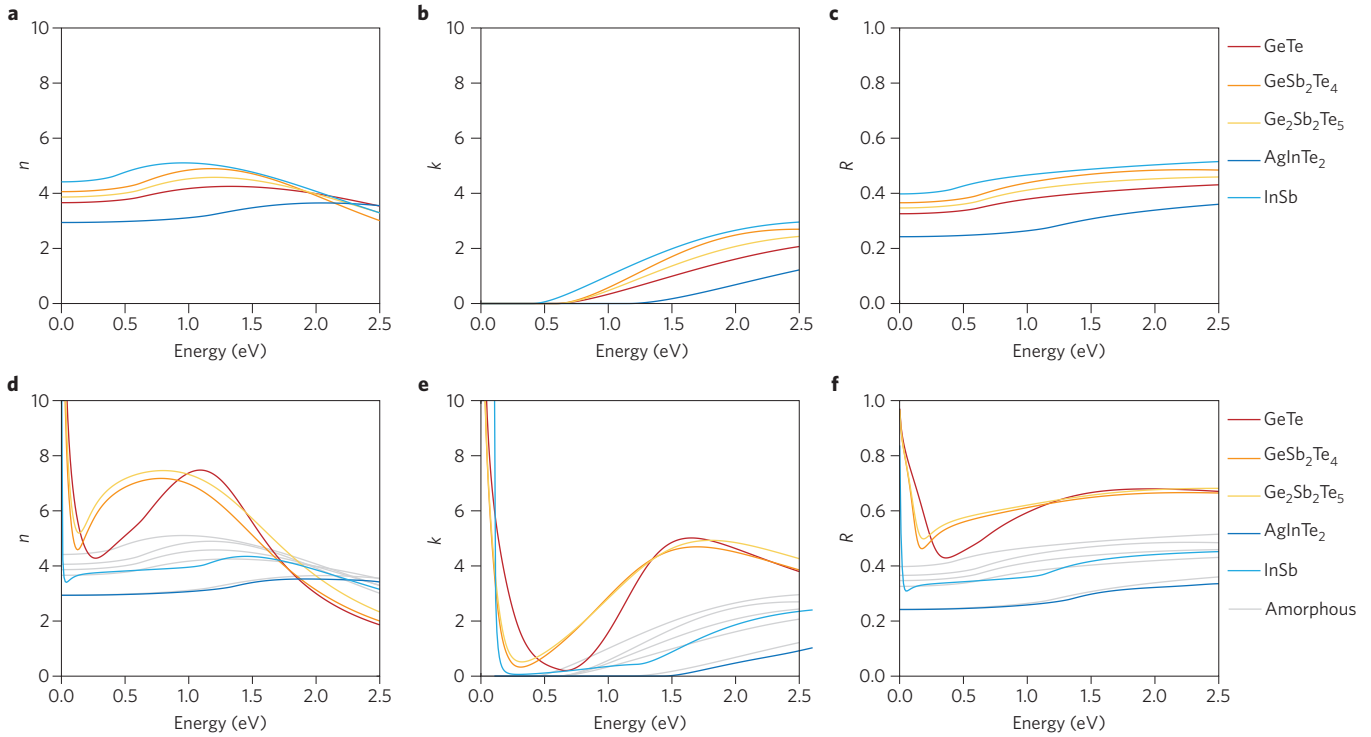

Figure 1 | Optical properties of PCMs. a–f, Refractive index (n), extinction coefficient (k) and reflectance (R) for three different PCMs (GeTe, $Ge_2Sb_2Te_5$ and $GeSb_2Te_4$) as well as two ordinary covalent semiconductors ($AgInTe_2$ and InSb and) as a function of photon energy for the amorphous (**a–c**) and crystalline (**d–f**) states. Only for the three PCMs, a pronounced change of optical properties is observed on crystallization, as can be seen by comparison with the grey lines for the amorphous phases. The increase in refractive index is accompanied by a decrease in bandgap in these materials. Finally, at very low energies, the contribution of free carriers to the dielectric function of the crystalline state, and hence the refractive index, extinction coefficient and reflectivity, is also visible.

excited, increases by more than 50% on crystallization for PCMs with resonant bonding. The changes in both bandgap and refractive index are best seen in reflectance spectra for thin films on metallic mirrors in ref. 21. We are not aware of any other class of materials that provides such a pronounced change of refractive index on transformation from the amorphous to the crystalline phase. Indeed, this increase cannot be explained by a concomitant increase in density like in ordinary semiconductors, but is rather due to a significant change in bonding, as discussed in Box 1. This bonding mechanism can only be realized for a subset of chalcogenides. As shown in Fig. 2, only a subset of chalcogenides feature resonant bonding where the bonding is not too ionic, as characterized by a significant electronegativity difference of the constituting elements, and there is not too much hybridization between the s and p valence electrons, which would lead to distortions of perfect octahedral coordination. Typical compounds that feature resonant bonding have bandgaps around 0.5 eV (≈2.5 μm). Even above the bandgap, the optical properties still differ significantly, but now pronounced absorption also takes place for both states. This can be detrimental for certain applications. The distinct optical contrast even extends down to the far infrared. As already mentioned above, the change in optical properties on the transition between the amorphous and crystalline states can be realized on the nanosecond and subnanosecond timescale. It is even possible to observe transient changes of optical properties on optical excitation[12,22].

The unique possibility to rapidly and drastically alter the optical properties of PCMs creates the foundation for emerging applications of PCMs in photonics. Depending on the desired photonic systems, different properties of PCMs are crucial, such as the property contrast, the minimum size of different optical states, the switching speed and so on. At present, there is significant progress or clear expectation for such advances in a number of different areas. The field of active photonics deals with the manipulation of light in photonic systems by external stimuli. Next we review the new concept of using PCMs for integrated optics and discuss how it can potentially fulfil the memory requirements for all-optical computing. Then, we move on to the field of nanophotonics, where we start with thin-film optics and colour displays and conclude with applications of PCMs for active plasmonics and metamaterials.

Integrated optics and all-photonic memories

A key challenge in overcoming current limitations of data processing and transfer is the speed of information transfer, not just over large distances, but also in systems and single devices. Chalcogenides and some PCMs have been used in fibre optics for almost three decades now[23]. However, it is only over the past decade that the field of on-chip photonics has seen significant progress. In particular, integrated waveguides and a range of photonic devices have been enabled by the proliferation of micro- and nanofabrication facilities. Information transfer by optical means is highly desirable because of almost unlimited bandwidth and the ability to multiplex. Yet, issues of residual cross-talk and the speed of photons make it very challenging to capture them in a compact device. Both computational interactions (that is, functions that carry out arithmetic operations, such as in a processor) and data storage are however essential components for a working photonic computer. Even in the short term, a limit to increasing computing speeds is the bottleneck in the shuttling of information between memory and processor in the so-called von Neumann computer. A solution would be to increase this bandwidth of information transfer on-chip by use of integrated optical circuits (or photonic circuits). A glaring limitation of photonic circuits is still the absence of high-quality, integrated photonic non-volatile memories. However, such a memory needs to be easily accessible in a chip-scale framework and thus requires full integration with the electronic circuitry. Particularly relevant are memory implementations that maintain

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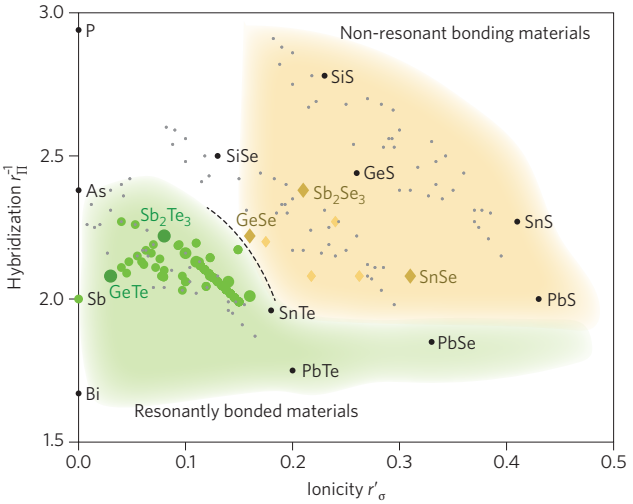

Figure 2 | Map for numerous compounds with an average of three *p* electrons. A wide variety of materials is shown that differ in their tendency towards hybridization between the valence *s* and *p* electrons ('covalency') and their ionicity. PCMs, marked by green circles, are all localized in the lower left corner of the map where ionicity and hybridization are both small. The boundary between materials with and without resonant bonding (yellow and brown diamonds), is denoted by a dashed black line. Materials that have not yet been characterized with respect to resonance bonding are denoted by black and small grey circles. Figure adapted from ref. 85, Macmillan Publishers Ltd.

their content once the driving power is switched off, that is, offer non-volatile data retention. To date, these contradicting requirements have been circumvented using only fairly complex implementations[13,14,24,25]. These, however, operate at timescales of mechanical resonances in the megahertz range and are far from current state-of-the art electronic memories that reach multi-gigahertz memory access rates.

This presents an important and highly relevant application for combining PCMs with integrated optical circuits for ultrafast and low-loss integrated optical functionalities. Materials used for integrated optics thus far overwhelmingly lack the option for active tunability. PCMs excel as potential candidates according to many recent demonstrations where they have been evanescently coupled onto nanophotonic circuits to build optical filters[9] and memory elements[26,27]. Most recently, it was shown that by evanescently coupling light onto $Ge_2Sb_2Te_5$ (GST-225, simply GST hereafter), a common PCM, the transmission could be altered by changing the state of GST between crystalline and amorphous phases[28]. In this embodiment, the state of GST governs the optical transmission in the waveguide; the transmission can be modified by choosing a higher-intensity pulse to change the phase. Thus, this provides a photonic analogue to electronic phase-change memories, where the current passing through a PCM cell enables either a readout or switching of the present state. The crucial difference in the implementation is that because GST is in itself a highly absorptive material, placing it directly in the path of light would attenuate the light significantly, increasing power consumption. Therefore, this implementation chose to couple the light evanescently to the PCM, by placing it on the waveguide and not directly in its path. The reason this works once again comes back to Fig. 1, that is, it involves a change in the complex refractive index of the PCM. As crystalline PCMs are 'incipient metals' (M. W., V. Deringer, X. Gonze, C. Bichara and J. Y. Raty, manuscript in preparation), they tend to be highly absorptive even below the bandgap[29], which results in a marked absorption of light. Any change in the solid phase alters the complex refractive index (both n and k), which affects the evanescent coupling to the PCM. As seen in Fig. 1, PCMs are characterized by a pronounced change in not only the refractive index n but also the extinction coefficient k on crystallization.

Using this technique, it has been possible to store and read data on a photonic chip. The first demonstration of such a memory has already described storage in (up to) eight levels instead of single bits, that is, eight different uniquely distinguishable transmission states[28] (Fig. 3). Crucially, the ability to arbitrarily move between any of the two levels within the system was demonstrated, which is an improvement over electronic phase-change memory capability. This by itself is a remarkable achievement, as deterministic and repeatable movement between levels is important for any long-term data storage application. This success represents the culmination of several decades of research into the material properties of PCMs and allows for the selective alteration of the crystalline fraction of the material. The crystalline fraction, that is, the ratio between the amorphous and crystalline volumes of the material, determines the effective complex refractive index. In turn, this determines the amount of attenuation within an optical waveguide. Thus, data is essentially stored in the state of the material. As these materials were originally engineered for data storage at acceptable power densities, the power required was shown to be of the order of only a few picojoules.

Although the technique is promising, it is not yet mature. It would require several additional improvements, the most important of which is a reduction in power requirement. Even if picojoules of power are reasonably competitive, this operation does not yet achieve the high sensitivity required for reliable multibit operations; for example, the lowest power demonstrated was only to store a single level. Multi-level operations require a higher signal-to-noise ratio, which in turn require higher-power devices. This one aspect needs significant improvement; thus, future research is crucial. Another key aspect of improvement would be in footprint reduction. At present, waveguides have a very large footprint (Fig. 3e), mostly because telecom wavelengths have been used, given the large-scale deployment of those wavelengths in modern technology. To lower the footprint, it is necessary to explore shorter wavelengths, which might move photonic memories outside of standard silicon and silicon nitride waveguides and possibly into aluminium nitride waveguides[30]. Yet, at these shorter wavelengths, PCMs tend to have a higher extinction coefficient.

In spite of these challenges, it is worth noting that the promise of this technology is immense and that alone warrants significant further research into the subject. For one, as the authors showed in the same paper[30], they were able to carry out multiplexing operations, being able to use multiple wavelengths simultaneously to carry out targeted read–write operations. This is at present not possible in electronic memories, and could potentially greatly increase data read and write speeds with cleverly designed algorithms. Second, with the von Neumann bottleneck dominated by the transfer of data between processor, memory and further into long-term data storage, the photonic transfer of this information would necessitate the use of photonic memories, and phase-change photonic memories provide a reliable and cost-effective solution. In future, the non-von Neumann processing techniques demonstrated in ref. 31 could be integrated into this all-photonic device, thus creating a photonic non-von Neumann processor.

Another important area where PCMs could contribute is on-chip routing, another essential component towards an all-optical architecture. Designs that combine novel routing mechanisms, for example, as proposed in refs 32–35, can potentially achieve this goal. A further size reduction might become possible by using switchable plasmonic techniques and waveguides[36], as explained in Box 2. Future developments in both areas are important for any practicable use of phase-change all-photonics memory elements or in future chips that require any processing.

Colour pixels, displays and smart glazing

The question that the discussion on photonic applications above left open was how switching in one domain (optical or electronic) changes the physical property of the other. Initial applications of

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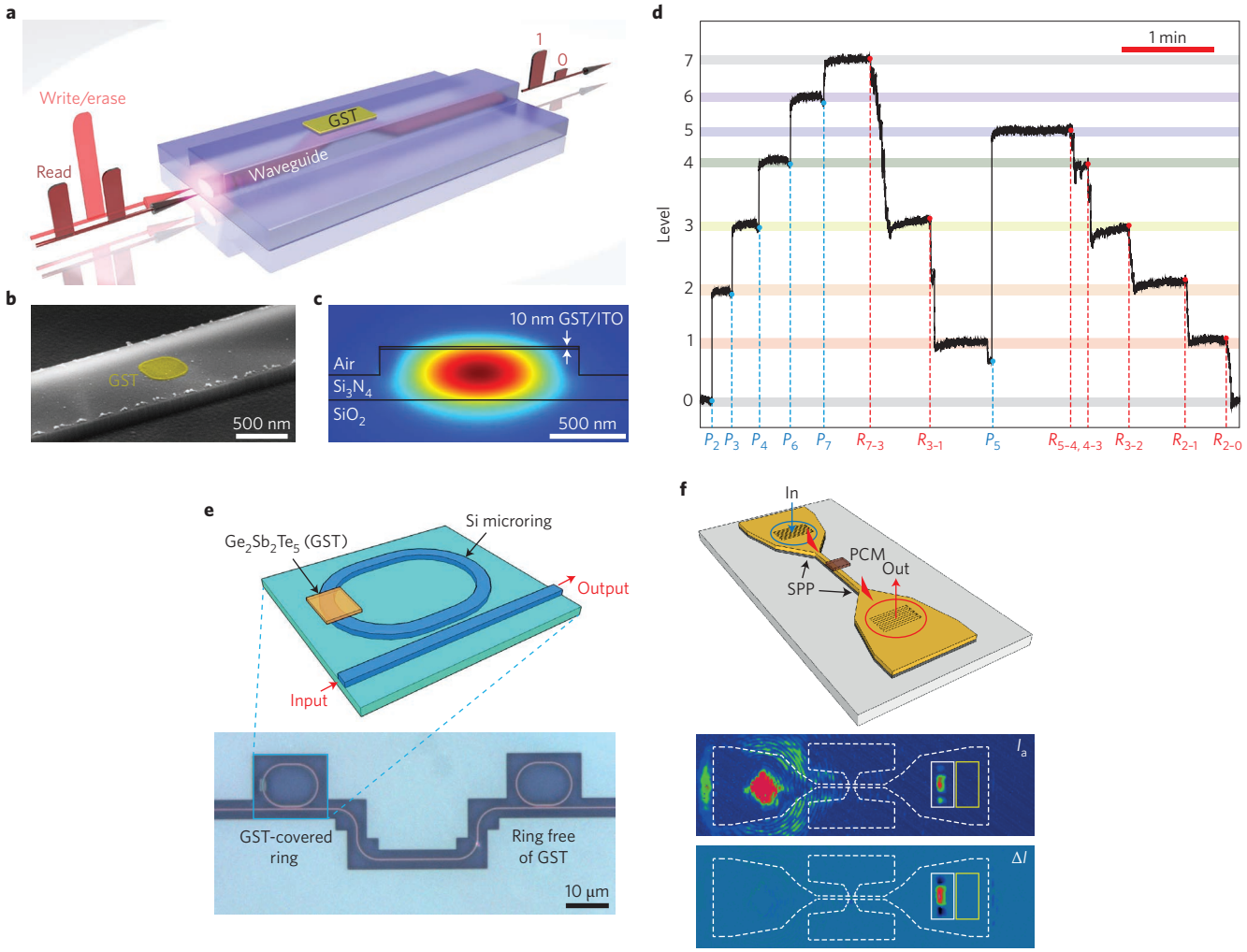

Figure 3 | Conceptual phase-change photonic memory. a,**b**, Sketch (**a**) and scanning electron microscopy image (**b**) showing how a PCM on top of a waveguide can be programmed to set and reset states and subsequently read out as a change in transmission of the waveguide. **c**, Simulated field distribution, which shows that the PCM is coupled evanescently to the light in the waveguide. ITO, indium tin oxide. **d**, Multi-level switching of eight transmission levels by applying write (P_i) and partial erase (R_{i-j}) pulses. **e**, Cross-section (bottom) and footprint (top) of a Si waveguide resonator employing a GST switch. **f**, Sketch of a miniaturized plasmonic waveguide (top), measured intensity distributions in the amorphpus state (I_a) (middle) and change of intensity (ΔI) (bottom) after crystallization of the PCM. Figure reproduced from: **a**–**d**, ref. 28, Macmillan Publishers Ltd; **e**, ref. 32, AIP Publishing LLC; **f**, ref. 36, American Chemical Society.

PCMs focused either on rewritable optical discs or on non-volatile phase-change memories based on the electrical conductivity change. Relatively few enquiries have been devoted to the mixed electro-optical properties of these materials. This is primarily due to the lack of mature applications. However, recent developments have created the need for such 'mixed mode' applications. By combining the electrical and optical properties together on nanoscale areas, new applications could emerge. This section reviews the most advanced of these in displays using thin-film optics concepts.

Before delving into applications, it is worth noting a restriction with the most commonly used nucleation-dominated PCMs that has so far impeded some possible applications. Early electronic switching studies already identified filamentation[37] as a significant issue in 'mixed mode' electro-optical applications. In this operation mode, on crystallization, a filament of highly conductive (crystalline) material forms, which prevents subsequent rapid crystallization of larger areas. This is detrimental for precise control of the crystalline fraction and hence the optical properties. Fortunately, there are also PCMs where crystallization is not dominated by nucleation but by crystal growth[38]. Apparently, the reduced glass transition temperature provides a first hint to the crystallization mechanism[39], with the lowest reduced glass transition temperatures favouring fast nucleation[40].

Thin-film optics has been an area of active recent interest, especially in the domain of colour rendering. This field was opened up by a paper by Kats *et al.*[41], where it was shown that using highly absorptive materials, such as metals, as very thin films can produce bright colours. The area of thin-film optics relies quite simply on the modulation of an optical cavity length with a stack of thin films. Thus, for example, in a simple thin-film stack, by changing the thickness of one or more layers one can tailor the resonance condition of the cavity. Another option is changing the film properties of one of the materials in the stack without changing film thicknesses. This is accomplished by changing the refractive index of that particular layer, which changes the optical resonance condition of the cavity. Hence, one can modify the wavelengths that are reflected out of a stack, enabling colour production of a certain hue in reflection.

The transfer matrix method[42] can be employed to obtain the reflection and transmission coefficients of the electromagnetic field in multilayered thin-film structures. Using PCMs as one of the thin-film layers provides an active functionality. This enables a modification

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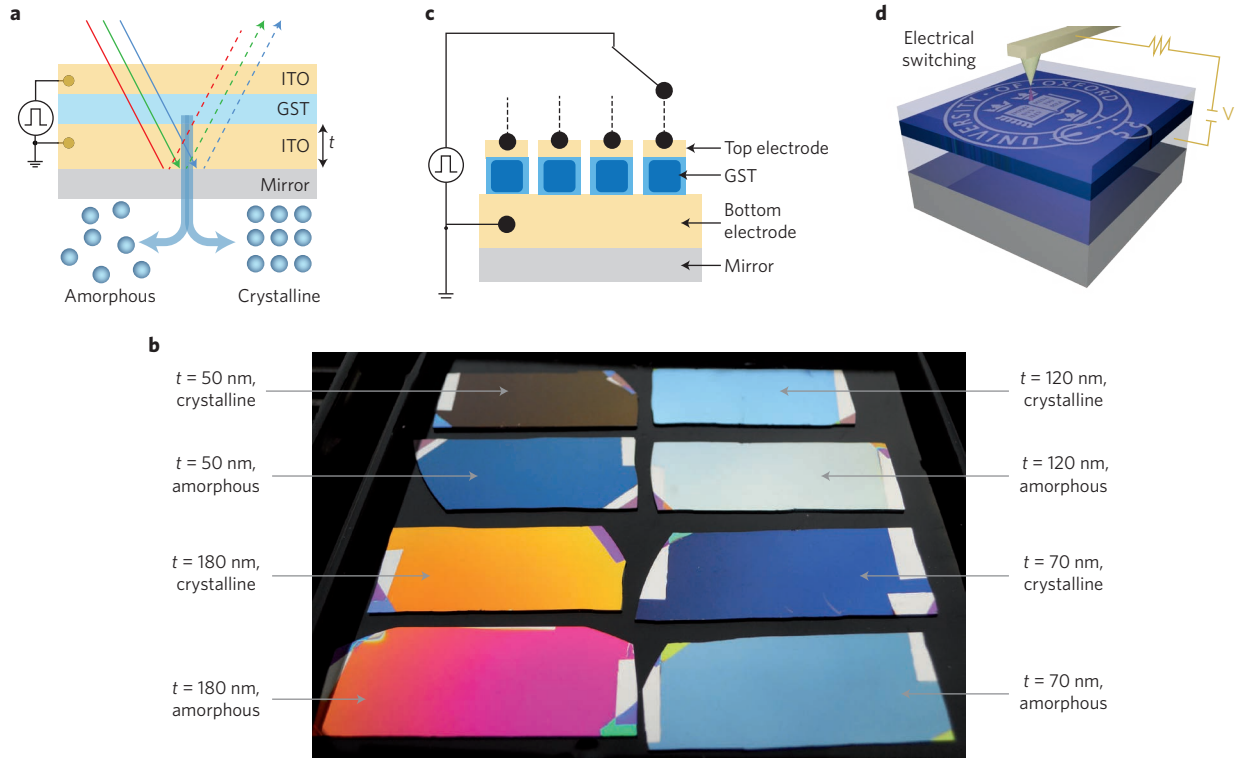

Figure 4 | Thin-film displays. a, Schematic representation of a thin-film layer stack that undergoes a colour change on crystallization of a PCM (here GST) when incident white light is partly reflected back (arrows). **b**, A variation of the PCM or spacer layer thickness *t* results in a broad range of colours. **c**,**d**, As PCMs possess the ability to retain the new state of refractive index indefinitely until changed again, bistable reflective displays can be created by local electrical switching. V, voltage. Figure reproduced from: **a**–**c**, ref. 43, Macmillan Publishers Ltd; **d**, ref. 44, Wiley.

of the refractive index on a structural transformation, which can be implemented by means of electric pulses. This enables true electro-optical operation of thin-film devices. Figure 4 describes the general concept of thin-film optics and the use of phase-change films.

Using these concepts, in 2014, Hosseini *et al.* reported the first nanopixel display producing stable colour changes in GST based on electrical stimulation[43]. This work describes an optoelectronic framework that exploits the amplification of optical reflectivity in a thin-film design, which employs an active layer of PCM. This film is sandwiched between two transparent conducting electrodes (indium tin oxide in this case) to create an optical cavity that could be addressed electrically. They also demonstrated cross-bar type pixels that could be reversibly switched. While any material that can undergo a change in its optical refractive index is applicable, PCMs are the most obvious candidate, due to their pronounced optical contrast, ability to be electrically switched and established industrial usage, that is, proven manufacturability. These successes warrant further enquiries to validate whether the concept is also suitable for larger scales and full display size arrays.

This aspect deserves further discussion. As pointed out earlier, a key limitation of PCMs in such applications (which was actually a key enabler for phase-change memories) is nucleation-dominated crystallization. When crystallization along filaments occurs, further crystallization requires a relatively large current. This current would increase exponentially as larger and larger areas require crystallization. Typical displays would employ crystalline pixels in the micrometre range or even above, much larger than the corresponding film thickness. It appears very demanding to switch such pixels energy efficiently. On the contrary, this facilitates applications that require exceptional resolution, as demonstrated in ref. 43, where pixel sizes of 50 nm were realized. Larger pixels would hence require growth-dominated PCMs.

Indeed, growth-dominated $Ag_3In_4Sb_{76}Te_{17}$ has been used to increase the crystalline area. Nevertheless, only marginally larger pixel sizes were obtained[44]. However, the applied voltages could better control the crystallization fraction. This resulted in partial crystallization, leading to intermediate values of the effective refractive index, enabling bistable pixels. Furthermore, in an analogue of multi-level phase-change memories, greyscale pixels were also possible. With similar techniques, but instead using a femtosecond laser-induced phase change, Schlich *et al.* reported another reflective display framework with a PCM on a thick layer whose colour can be fully switched[45]. They also demonstrated that such colour is robust with respect to the angle of incident light.

The outlook for this application of PCMs is promising. Phase-change nanodisplays have significant advantages because they use ultrathin films. The first is that these materials are already qualified for use in electronic applications, rendering their commercialization more feasible. Thin-film deposition techniques are well established and do not require new processes to be developed. Importantly, such thin films can be deposited on any substrate, including thin flexible films, as the process is inherently low temperature[43]. This could enable future flexible and rollable displays. Furthermore, it was shown recently[46] that the colour gamut of these displays can match or exceed that of organic light-emitting diodes, thus promising an interesting application of these remarkable materials.

Looking ahead, it is worth noting (Fig. 1) that the change in refractive index of such materials is even more pronounced in the near infrared (a feature exploited in the photonic memories described in the previous section). This promises a range of exciting applications, such as truly smart windows that primarily modulate the invisible spectrum of solar radiation, maintaining ambient conditions indoors, or displays integrated with smart windows on a large scale. One can even hope to use PCMs in future virtual reality and augmented reality devices to create (holographic) images and videos. To explore the full potential of this class of materials, we also need to stress present limitations of the materials. These are related to the power necessary to switch, the switching speed and the cyclability, that is, the number

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of possible switching cycles before degradation takes place. This is closely linked to the design of the device and hence the compounds surrounding the PCM. More than one million cycles have been realized in rewritable optical disks and more than one billion cycles in non-volatile electronic memories[11].

Active plasmonics and metamaterials

The integrated photonic applications of colour displays mentioned above already exploit very thin films for the generation of different colours. In this section, we focus on lateral nanostructures and the potential to create and utilize such structures. This will result in exciting functionalities for optical elements enabled by means of 'plasmonic' nanostructures and arrays of them, so called 'metamaterials' (Box 2). Focusing and guiding of light on length scales below the diffraction limit is possible by exploiting the strong field confinement and enhancement close to these metallic surfaces and nanostructures. Plasmonic waveguides, for example, promise a smaller footprint compared with dielectric waveguides. Already in 2004, the term 'active plasmonics' was introduced in a paper reporting on the concept of controlling propagating surface plasmon polaritons (SPPs) by using optically activated phase transitions in metals[47]. Later on, the same group also first proposed and demonstrated the use of chalcogenide glasses in active plasmonics[48,49]. The idea of putting a patch of PCM next to a plasmonic waveguide, similar to the concept shown above for dielectric waveguides, was first realized in 2015 by Rudé *et al.*[36]. They were able to switch optically a small patch of GST and thus control the waveguide's transmission (Fig. 3f).

Plasmonic nanostructures can also be made entirely from a PCM, at least at frequencies below the plasma frequency. The Drude contribution in the crystalline state can lead to a negative permittivity. Polking *et al.* used this behaviour to control the localized surface plasmon resonance in GeTe nanoparticles that emerges on crystallization[50]. In contrast, PCMs have been used for visualizing the locally enhanced near-fields of resonant structures. The locally enhanced fields of SiO_2 microspheres have been imprinted in GST and studied with optical, electron and atomic force microscopy. Erasable features with sizes down to 10 nm have been written[51,52].

Tailoring artificial optical materials with plasmonics, that is, metallic resonators, has led to a multitude of new device concepts for waveguides, beam shaping and focusing, polarization control, holography, sensing and imaging[53,54] (Box 2). Commonly, the optical properties of most plasmonic and metamaterial structures are limited after fabrication to a certain functionality or operating wavelength. However, optical components with either optically or electrically controllable parameters have enormous technological importance. The reversibly switchable optical properties of PCMs, that is, the huge change in refractive index in combination with the at least two stable states, render them a key ingredient for non-volatile, active nanophotonics. If a PCM is placed next to a resonant nanostructure, the change in its refractive index will shift the structure's resonance and give rise to altered functionalities, for example, a change of polarization state of the light, a change of the focal length of a lens or a change of the target wavelength of a detector for thermal radiation.

Since PCMs are mainly used in active plasmonics and active metamaterials as a switchable dielectric, that is, as a surrounding medium, multiple considerations and requirements for the active material have to be met.

The first decision when searching for materials for active plasmonics strongly depends on the desired application. Is a volatile change desired, that is, the change only occurs while the external stimulus is turned on (for example, for a modulator)? Alternatively, does one want to achieve a reconfigurable functionality that is non-volatile, that is, keeps the switched state even after the external stimulus is turned off? While the first route allows many interesting concepts — such as electrical or optical carrier injection/depletion in semiconductors or temperature-driven phase transitions between insulating and metallic states as observed, for example, in vanadium dioxide[55], which was recently also shown to become non-volatile when prepared under distinctive deposition conditions[8] — it is clear that the main advantage of using PCMs for active photonics is the well-established, non-volatile amorphous and crystalline phases.

The next important requirement is a very pronounced optical contrast (refractive index change) between the amorphous and crystalline phases in the spectral range of interest to create large resonance shifts. This is where PCMs also excel due to their resonant bonding mechanism in the near- and mid-infrared spectral range (Fig. 1). The initial experiments on switching metamaterial resonances in the near-infrared spectral range in 2010 used a thin layer of the exotic chalcogenide glass gallium lanthanium sulfide (GLS) to shift the absorption dip of a metamaterial resonance[49]. The metamaterial consisted of asymmetric split-ring resonators coupled to slits, which are both milled via focused ion-beam etching into a metallic film. This special configuration of two entities within the metamaterial unit cell enables an especially narrow resonance, a so-called Fano resonance[56], due to the interaction of dark and bright modes. A couple of years later, the more common PCM GST was exploited for stronger resonance shifts in the near- and mid-infrared spectral range. Zheludev's group improved their pioneering 2010 design[49] by sandwiching a thin GST layer in between ZnS/SiO_2 layers, which acted as buffer and capping layers, respectively, and placing this sandwich on top of a gold metamaterial[57] (Fig. 5a). They showed that a (rather broad) metamaterial resonance could be optically switched by single 660-nm laser pulses with diameters of about 50 μm, yielding transmission contrast ratios of 4:1, similar to their previous work based on GLS[49]. Other groups used a 20-nm-thin GST layer sandwiched between a gold nanodisc array and a quartz substrate and achieved non-volatile tuning of the lattice resonance from 1.89 to 2.27 μm wavelength by thermal crystallization of GST[58] (Fig. 5c). They also investigated the evolution of the resonance shift for different baking times and modelled this 'continuous' tuning with partially crystallized GST.

Another important requirement for many desired designs of active metamaterials is the absence of losses. In the spectral range between the bandgap absorption and the optical phonons at low frequencies, amorphous PCMs show no optical absorption and thus have a zero imaginary part of the permittivity. For the crystalline state, absorption due to free charge carriers is present at low frequencies. Thus, for each material there is a specific 'window of low losses' between the optical bandgap and the Drude absorption. For $Ge_3Sb_2Te_6$ (GST-326), this leads to lower losses in the mid-infrared frequency range compared with the commonly used GST, as pointed out by Michel *et al.*[59] (Fig. 5b). GST-326 exhibits a very large difference in the real part of the permittivity (ε) between the amorphous (a-) and the crystalline (c-) phases within a broad mid-infrared wavelength region ($\varepsilon_{a\text{-}GST} \approx 12 + 0i$, $\varepsilon_{c\text{-}GST} \approx 36 + 2.5i$ at least from 800 to 2,500 cm^{-1}), while the imaginary part of both phases is comparatively small ($Im(\varepsilon) / Re(\varepsilon) < 0.1$). To maximize the interaction with the near fields of the rod-shaped infrared antennae, spacer layers were omitted and aluminium was chosen as an antenna material showing less diffusion into the PCM compared with gold. Using the PCM GST-326, they achieved a maximum resonance shift of 19.3% to lower frequencies and a 'tuning figure of merit', defined as the resonance shift divided by the full-width at half-maximum of the resonance peak of 1.03. They also achieved a shift to higher frequencies using the semiconductor material InSb, which exhibits the counterintuitive property of decreased density (and refractive index) on crystallization[60]. However, the thermal switching on a hot plate restricted their experiments into one direction, from the amorphous to the crystalline state.

The next requirement is the possibility to switch reversibly between the different states. Endurance (that is, the number of possible switching cycles), a high switching speed (given by the crystallization step) and a low energy consumption are critical issues to be solved before commercial applications will arise. Regarding

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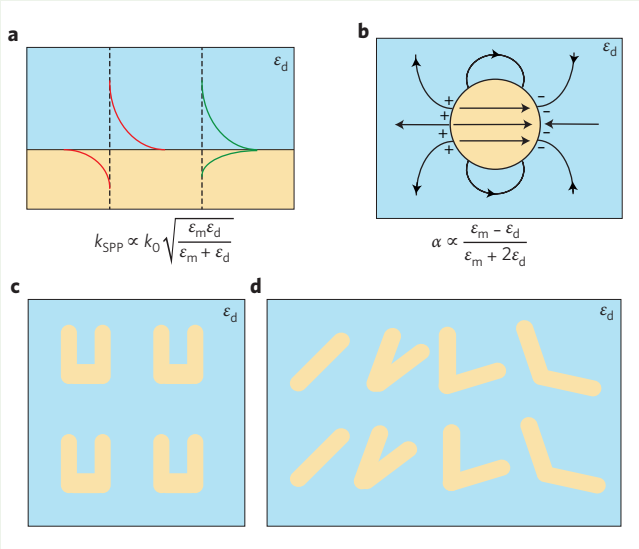
Box 2 | Introduction to plasmonics and metamaterials.

The term 'nanophotonics' describes photonics below the diffraction limit of light, that is, where at least one length scale of the structures involved is below one half of the wavelength in the surrounding medium (λ / ($2n$)). The focusing, waveguiding and manipulation of light on a subwavelength scale becomes possible by using metallic surfaces and nanostructures, which enable local field confinement and strong field enhancement. Two main directions of nanophotonics research currently address this subwavelength manipulation of light: plasmonics and metamaterials.

Plasmonics deals with the optics of metallic nanostructures and surfaces, where collective charge oscillations in metal surfaces and nanoparticles ('plasmons') are coupled to light and thus create hybrid modes (for example, a 'plasmon polariton') that can have much greater confinement of the (electric) light field and a much shorter wavelength compared with photonic waveguides[53,92,93]. Typical implementations rely on propagating SPPs (panel a), which are confined at the surface and propagate along it with short polariton wavelengths (the surface plasmon polariton wavevector $k_{SPP} > k_0$, where k_0 is the wavevector in free space, depends on the permittivitty of the metal, ε_m, and of the surrounding dielectric, ε_d). Another implementation relies on localized surface plasmon resonances in small metallic nanoparticles (panel b), where the particle's polarizabiliy α also depends on ε_m and ε_d. Even in extended nanostructures made of perfect metals, for example, at infrared frequencies, geometric resonances, for example, of rod-like λ / ($2n$) dipole antennae exist. All of these configurations create a strong optical field enhancement and confinement, especially at their resonance frequency. The resonances in all cases shown depend also on the permittivity ε_d of the dielectric environment, which can be altered using PCMs.

Metamaterials are artificial materials that are structured on a subwavelength scale and can be described by an effective medium approach[94–97]. For optical metamaterials, the unit cell often consists of metallic nanostructures with electric and magnetic resonances (panel c). By tailoring the latter, it is also possible to influence the light fields not only via the permittivity ε_{eff}, but also via the magnetic permeability (μ_{eff}) at optical frequencies, something not possible with conventional materials. Thus, new optical concepts, such as negative refraction or subwavelength imaging with perfect lenses, can be explored.

Both plasmonic nanostructures and metamaterials allow for the creation of arbitrary light field distributions on a subwavelength scale. The concept of so-called metasurfaces[98] (panel d) exploits this flexibility for changing the propagation direction, polarization state,



spatial amplitude, phase patterns and so on of the incident light in an arbitrary way. Key to the creation of arbitrary spatially varying light fields is the local, spatially varying adjustment of amplitude and phase of an incoming light beam by the individual metallic nanostructures. It has to be noted that for metasurfaces, the phase shift of the reflected or transmitted light is created by resonant elements of subwavelength thickness and not accumulated by propagation, like in conventional phase plates.

Common to most plasmonic and metamaterial structures is that after fabrication the optical properties are limited to a certain functionality or operating wavelength. However, optical components with adjustable parameters have enormous technological importance. The creation of such adjustable functionalities with nanostructures is the aim of the research field called 'active plasmonics'[99] where external stimuli such as heat, stress, electrical fields or optical pulses are used to modify the nanostructures' optical response. In many of the investigated systems, the new state only remains while the external stimulus, such as strain[100], electrical voltage[66,101–104], optical pulses[64] or temperature[55], is turned on. Thus, all these mentioned concepts are limited to volatile applications. The reversibly switchable optical properties of PCMs, however, enable active, nonvolatile nanophotonics.

the number of switching cycles of active metamaterials and plasmonic structures, a lot of improvement is possible before we reach the values demonstrated for optical data storage with PCMs. While the authors in ref. 57 claim an endurance of about 50 switching cycles, unfortunately no data on the reversible switching were presented. Here, the diameter for optical switching was restricted to about 50 μm, while other authors did show large-area reversible switching of the mid-infrared antenna resonances using femtosecond laser radiation[61]. Here, a single 50-fs laser pulse with a relatively high fluence of 51 mJ cm^{-2} was used for amorphization of a whole antenna array with laser spot sizes on the order of 250 μm diameter. For crystallization, a sequence of pulses with a lower fluence of 31 mJ cm^{-2} was employed, yielding a slightly lower redshift of the resonance compared with the first, thermal crystallization. In the same year, a theoretical paper on a broadband perfect absorber at visible frequencies was published[62]. Due to their high absorbance, these structures are predicted to enable a crystallization of the GST film at very low intensities of 95 nW μm^{-2}.

The fourth important requirement and advantage of PCMs for active plasmonics and metamaterials is that the switching can be induced in a flexible way by various means, that is, by local heating, optically or electrically. The initial demonstration of active metamaterials using GLS[49] in 2010 used electrical currents to achieve the crystallization of the GLS with electrical pulses of 10 ms lengths and increasing voltage. This electrical excitation was possible as the metamaterial consisted of asymmetric split-ring resonators coupled to slits, which were both milled via focused ion-beam etching into a metallic film. Reversible electrical switching was not demonstrated in this publication. As mentioned above, when discussing the colour displays, the size of the electrically switched area will be limited due to filamentation. However, by clever design of the metamaterial unit cell, it might be possible to obtain a huge resonance shift by only switching a small amount of material within a small gap between two arms of two antennae[63] (or within the gap of a split-ring resonator). This has been reported for optically excited active antennae based on the (volatile) injection of free carriers in doped semiconductors[64]. Electrically

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switchable metamaterials, based on doped semiconductors, have already been demonstrated in the terahertz and mid-infrared spectral range by applying a common bias voltage over all structures[65,66] simultaneously. In all demonstrations shown so far, all structures were altered the same way and no local addressing of individual elements was shown. Such an electrical, local addressing of individual metamaterial unit cells would require a huge amount of contacting wires, which might interfere with the metamaterials' optical properties. In contrast, the use of PCMs for individually addressable metamaterials might benefit from the development of a highly integrated, fully addressable phase-change random access memory[67,68] or the colour display presented above. If electrical contacts already exist for individual cells, a clever design could potentially make them act as 'plasmonic' structures as well. Another option for reversible electrical switching of the PCM over larger areas has recently been demonstrated for radiofrequency switches, where a line heater (metal wire) was used to crystallize and amorphize GeTe over an area of up to 2 μm × 18 μm via proper voltage pulses and thermal design[69,70]. Due to the huge efforts needed for large area or addressable electrical switching of PCM metamaterials, the optical switching and addressing seems more accessible at present.

After the discussion of fundamental requirements for active plasmonics and metamaterials and how PCMs can fulfil these, we now present and discuss the functionalities that can be enabled. As mentioned before, in general, the use of PCMs allows for wavelength tuning of the desired functionality. This has been initially used for changing the operating wavelength of negative index metamaterials and antenna arrays, but also for perfect absorbers of devices that change the polarization state of the light.

Active metamaterials using PCMs

After the initial experiments on PCMs and related materials as a switching agent for active metamaterials, many theoretical proposals and electromagnetic simulations soon followed. A tunable negative index metamaterial based on elliptical nanoholes in two 30-nm-thin metallic gold films with a 30-nm-thin layer of $Ge_2Sb_1Te_4$ (GST-214) in between as a switchable dielectric was proposed in ref. 71. A tunable, polarization-independent 'perfect' absorber of infrared radiation with more than 80% absorption and a 10% tuning of the absorption peak based on metallic square patches on a thin GST film above a reflecting gold plane was proposed in refs 72 and 73. The same group also proposed a design using chiral metamaterial structures combined with a thin GST layer to demonstrate tunable circular dichroism, that is, a change in the spectral position where light with different circular polarization (left-handed and right-handed circularly polarized light) is transmitted or absorbs differently[74].

Recently, exciting new functionalities of PCM-based active metamaterials have been realized. Chen *et al.* reported a switchable plasmonic lens based on slits filled with GST in a gold film[75]. Depending on the level of crystallization, the Fabry–Perot resonance in each slit can be shifted spectrally. This way, the phase of the transmitted light at the operating wavelength of 1.55 μm can be adjusted in theory over a 0.56π range. In practice, the authors only achieved full crystallization at certain slits by scanning a visible laser along them, thus creating a binary optical phase pattern. The experimental results do indicate changes in focal lengths or propagation direction, but the binary pattern reduces the beam quality.

In 2015, Yin *et al.* used stacked metallic nanorods in two enantiomer configurations with a layer of PCM in between to demonstrate the first switching of a chiral metamaterial[76] (Fig. 5d). Exploiting the extraordinary properties of GST-326, namely the huge refractive index change and the low losses, they were able to tune the circular dichroism response from 4.14 to 4.9 μm wavelength. The circular dichroism is defined as the difference in the transmittance of right-handed and left-handed circularly polarized light. Furthermore, by adding another layer acting as a 'bias layer' for the chirality, they even demonstrated a reversal of the sign of the circular dichroism on PCM crystallization at the desired design wavelength.

The first experimental demonstration of a switchable perfect absorber with application potential for thermal imaging was achieved in the same year by the same groups[77]. They also used a thin GST-326 film in between a reflecting aluminium mirror and square aluminium nanopatches to shift the wavelength of almost perfect absorption (>90%) by a huge amount, from 2.95 to 3.6 μm (Fig. 5e). Following the thermal emission via Planck's law, a narrowband perfect absorber corresponds to different temperature ranges of the emitter. Thus, by fabricating aluminium patches of different sizes, an application example would be thermal imaging with an adjustable peak detection temperature range.

Also in 2015, two theoretical studies by Cao *et al.* addressed advanced functionalities of PCM-based active plasmonic structures[78,79]. Using stripes of metal–insulator–metal structures with GST as a switchable insulator in between shifted the plasmonic resonance and, with proper design of the geometry, beam steering was proposed[78]. In the second publication, the author proposed the creation of circular conversion dichroism under oblique incidence using an achiral structure (square patches of gold above a GST layer and a gold reflector). A huge frequency shift of up to 45% is predicted for the circular conversion dichroism resonance based on a magnetic dipolar resonance in the multilayer structure[79].

Perspectives

So far, only a few basic concepts using PCMs for active plasmonics and metamaterials have been proposed, and even fewer have been experimentally realized. Most of the concepts only demonstrated switching from the amorphous to the crystalline state. Fully reversible optical or electrical switching is more elaborate, but, as pointed out earlier, is clearly within reach. While the future possibilities of switchable metallic nanostructures with PCMs are countless and many structures with different functionalities, such as filters, lenses, absorbers, sensors and so on, can be imagined, in all of the presented concepts for active nanophotonics, metallic nanostructures are employed for the creation of subwavelength resonators. However, the use of metals always comes with the inherent losses associated with them, and the nanostructuring of metals demands multiple, cumbersome fabrication steps. After fabrication, the optical properties are commonly limited to a single functionality. The present designs are also restricted to large-area switching, that is, all nanostructures undergo the same change when being switched. All of these factors hamper the practical use of active plasmonic structures. Thus, there will be a need for new concepts and materials, for example, all-dielectric metamaterials[80], to overcome these limitations and to implement new functionalities. The independent control of each individual nanostructure's optical response will ultimately lead to the vision of 'metasurfaces on demand'[81]. The first experimental proof-of-concepts for this exciting direction were published independently by two groups in 2016.

Zheludev's group again pioneered this field by demonstrating all-dielectric optically reconfigurable photonic devices and metasurfaces based on PCMs[35,82]. In ref. 35, they used trains of diffraction-limited optical femtosecond laser pulses to locally write patterns with a continuously adjustable reflectance change in a GST film sandwiched between ZnS/SiO_2 layers on a glass substrate. These patterns can be written and erased (Fig. 5f). A variety of devices have been demonstrated, ranging from reconfigurable, bichromatic and multi-focus Fresnel zone plates, a super-oscillatory lens pattern, a greyscale hologram and a 'metamaterial' consisting of dielectric resonators. The absence of metals reduces the losses in the resonators, but also limits the local field enhancement and requires sizes of the resonators on the order of the wavelength divided by the refractive index (λ / n). Another interesting, recently published approach replaces the metals with a polar crystal supporting surface phonon-polaritons (SPhPs), the infrared counterpart of SPPs.

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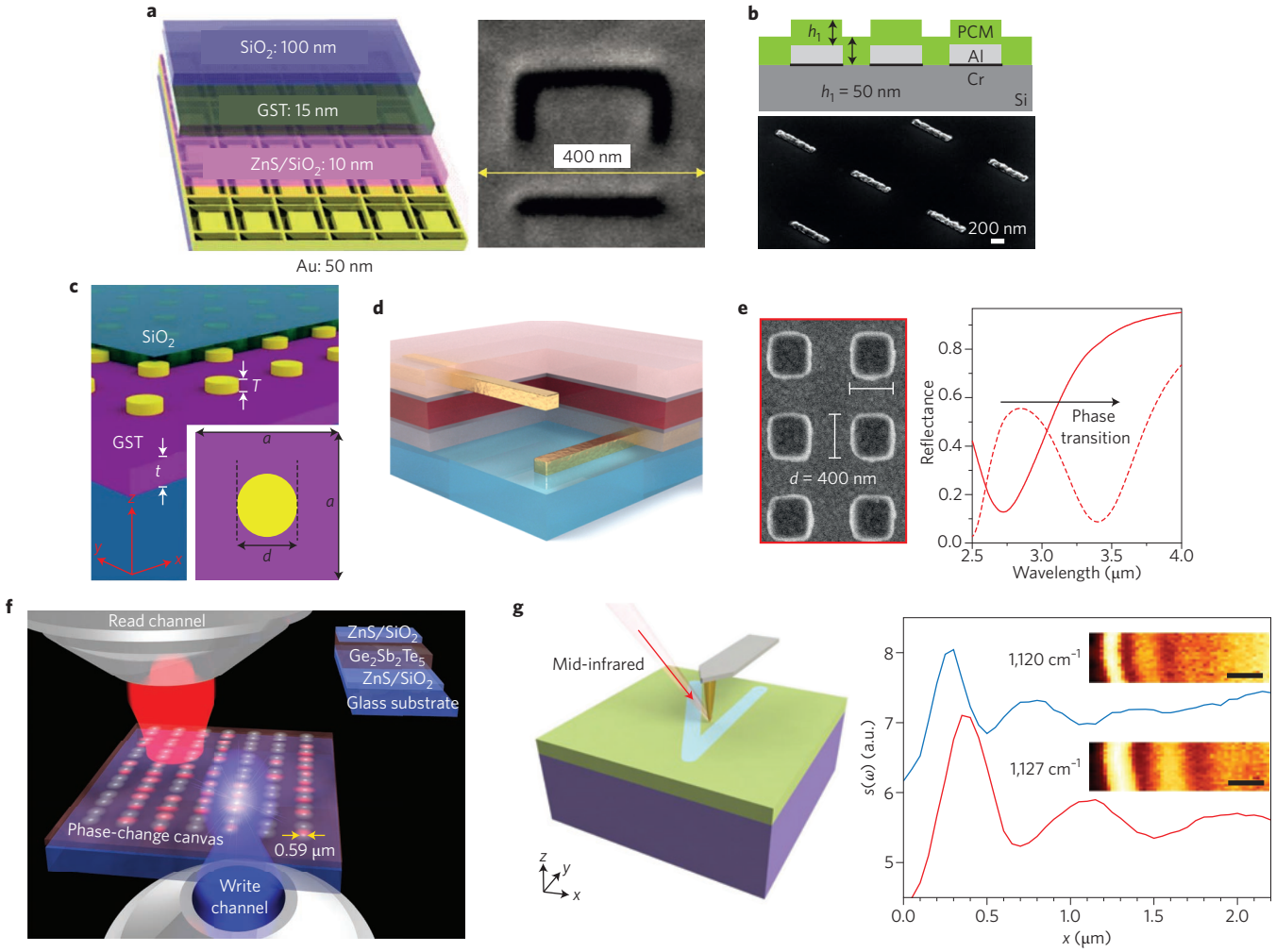

Figure 5 | PCMs for active nanophotonics. a, Sample layout (left) and scanning electron microscopy (SEM) image (right) of the unit cell used for the optical switching of a metamaterial resonance using GST. **b**, Sample layout (top) and SEM image (bottom) for low-loss infrared antenna resonance switching using a thin film with height h_1 of the PCM GST-326 for a red-shift and InSb for a blue-shift on crystallization while achieving a high-resonance quality factor. **c**, Sketch of an array of gold dots (period *a*, diameter *d* and height *T*) on a thin GST layer (thickness *t*) used for shifting of a lattice resonance (period *a*). **d**, Sketch of basic structure for switchable circular dichroism consisting of two stacked gold rods on a transparent substrate (blue) embedded in a capping layer (white) with a thin PCM layer (red) in between. With a more elaborate design even the change the sign of the circular dichroism is possible on crystallization of the PCM (GST-326). **e**, Switchable perfect infrared absorber using GST-326. SEM image (left) and experimental reflection spectra (right) show a switchable narrowband infrared absorption (solid line, amorphous GST; dashed line, crystalline GST). **f**, Towards rewritable metasurfaces. Optical writing and reading of dielectric resonators into a thin PCM film in arbitrary patterns that can act as lenses or holographic patterns. **g**, Left: SPhPs with ultrashort polariton wavelengths (λ / 50) on a polar crystal covered with a thin PCM film after laser-based structuring (PCM switching), visualized with a scanning near-field optical microscope. Right: line profiles of the optical amplitude $s(\omega)$ at two different frequencies (blue and red) prove the existence of ultra-confined SPhPs. Scale bars, 500 nm. The examples in **f** and **g** show a new direction for active nanophotonics: getting rid of the lossy metals and enabling direct optical writing of functional metamaterials on demand. Figure adapted from: **a**, ref. 57, Wiley; **b**, ref. 59, American Chemical Society; **c**, ref. 58, OSA; **d**, ref. 76, American Chemical Society; **e**, ref. 77, Wiley; **f**, ref. 35, Macmillan Publishers Ltd; **g**, ref. 83, Macmillan Publishers Ltd.

Li *et al.* demonstrated that an ultrathin layer of GST-326 can dramatically confine SPhPs on a quartz surface (Fig. 5g), while the local switching of the PCM can create subwavelength resonators for the SPhPs[83]. As mentioned in the related Commentary[84], their approach in principle allows for changing the functionality of the structure from scratch in a non-transient fashion and represents a genuinely rewritable platform for subwavelength optical components.

With the platforms provided in these two publications, functional optical elements can now simply be written optically in thin PCM films without the need for cumbersome structuring of metallic nanostructures. PCM-based photonic elements can rely on the well-established platform of optical data storage to create and erase optical functional elements on demand. The writing process can be accelerated using fast laser scanner and pattern generators. Proper layer and materials design, as previously done for optical data storage, can improve the number of switching cycles. The solid understanding of the physics of PCMs will enable PCM properties to be tailored to the desired application or spectral range. Depending on whether a large contrast in the real or imaginary part of the permittivity is required, a red- or blueshift on crystallization is needed or a change of sign in the permittivity or low losses are desired, distinct PCM compositions can be employed. Thus, we see a bright future for 'photonic devices on demand' using the extraordinary properties of PCMs.

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Competing financial interests

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